

STABILITY/MOTOR CONTROL ASSESSMENT



Rolling Tests are used in the SFMA to grade the level of severity of any stability/motor control dysfunction. This test helps identify if the stability/motor control dysfunction is a local problem or more of a global problem. If rolling is dysfunctional, we know there is a fundamental level dysfunction.

It is appropriate to use Rolling Tests in any breakouts with findings of stability/motor control dysfunctions. Here are examples of how Rolling Tests are used in the four main breakouts:

Multi-Segmental Flexion

If the Rolling Test is needed within the Multi-Segmental Flexion pattern, you must test all four flexion rolling patterns:

- Supine to Prone Upper Body Rolling (both arms)
- Supine to Prone Lower Body Rolling (both legs)

Multi-Segmental Extension

If the Rolling Test is needed within the Multi-Segmental Extension pattern, you must test all four extension rolling patterns:

- Prone to Supine Upper Body Rolling (both arms)
- Prone to Supine Lower Body Rolling (both legs)

Multi-Segmental Rotation

If the Rolling Test is needed within the Multi-Segmental Rotation pattern, you must test all eight rolling patterns:

- Supine to Prone Upper Body Rolling (both arms)
- Supine to Prone Lower Body Rolling (both legs)
- Prone to Supine Upper Body Rolling (both arms)
- Prone to Supine Lower Body Rolling (both legs)

Single-Leg Stance

If the Rolling Test is needed within the Single-Leg Stance pattern, you must test all eight rolling patterns:

- Supine to Prone Upper Body Rolling (both arms)
- Supine to Prone Lower Body Rolling (both legs)
- Prone to Supine Upper Body Rolling (both arms)
- Prone to Supine Lower Body Rolling (both legs)

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How to Interpret Rolling Results:

When performing rolling from supine to prone (flexion pattern rolling) there are several compensatory movements that make the test dysfunctional:

- Going into extension. This is a flexion type pattern and the patient should not recruit an extension pattern. This includes kicking a foot into the ground, arching their back, extending their head, etc.
- Simultaneously flexing the upper and lower extremities is a sign of dysfunction.
- Log rolling - when the upper and lower bodies land at the same time.

When performing rolling from prone to supine (extension pattern rolling) there are several compensatory movements that make the test dysfunctional:

- Going into flexion. This is an extension type pattern and the patient should not recruit a flexion pattern. This includes pressing a foot or arm into the ground, flexing the hip, flexing the neck, etc.
- Simultaneously extending the upper and lower extremities is a sign of dysfunction.
- Log rolling - when the upper and lower bodies land at the same time.

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Test Objective for the Supine to Prone Upper Body Rolling Test:

This tests the ability to activate and sequence the local stabilizers for upper body flexion pattern.

How to Perform the Supine to Prone Upper Body Rolling Test:

Instruct the patient to lie supine, on a flat surface with the legs extended and the arms flexed overhead. While keeping their lower body relaxed, they should reach their right arm up and reach across their body, while also flexing their neck, to roll their body onto their stomach. Repeat using the left arm.

Functional Criteria:

Normal Supine to Prone Upper Body Rolling is the ability to roll from their back to their stomach, without using their lower body, in a controlled manner.

Additional Information:

Many times, gross stability dysfunctions combined with flexion maneuvers can be identified using rolling patterns. A good sequence of core stability and segmental loading of the entire body is required to perform this test correctly.

Watch for simultaneous flexion from the lower body, log rolling, and/or using extension to complete the pattern.

Possible Findings:

- Functional: Ability to complete supine to prone upper body rolling, no pain
 - » Indicates a higher postural stability/motor control dysfunction for upper body flexion sequencing.
- Dysfunctional: Inability to complete supine to prone upper body rolling, no pain
 - » Indicates a fundamental level stability/motor control dysfunction for upper body flexion sequencing.

*If there is pain with the rolling pattern, consider it dysfunctional and remain in the Supported posture.



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Test Objective for the Supine to Prone Lower Body Rolling Test:

This tests the ability to activate and sequence the local stabilizers for lower body flexion pattern.

How to Perform the Supine to Prone Rolling Lower Body Test:

Instruct the patient to lie supine, on a flat surface with the legs extended and the arms flexed overhead. While keeping their upper body relaxed, they should lift their right leg up across their body to roll onto their stomach. Repeat using the left leg.

Functional Criteria:

Normal Supine to Prone Lower Body Rolling is the ability to roll from their back to their stomach, without using their upper body, in a controlled manner.

Additional Information:

Often, gross stability dysfunctions combined with flexion maneuvers can be identified using rolling patterns. A good sequence of core stability and segmental loading of the entire body is required to perform this test acceptably.

Watch for simultaneous flexion from the upper body, log rolling, and/or using extension to complete the pattern.

Possible Findings:

- Functional: Ability to complete supine to prone lower body rolling, no pain
 - » Indicates a higher postural stability/motor control dysfunction for lower body flexion sequencing.
- Dysfunctional: Inability to complete supine to prone lower body rolling, no pain
 - » Indicates a fundamental level stability/motor control dysfunction for lower body flexion sequencing.

*If there is pain with the rolling pattern, consider it dysfunctional and remain in the Supported posture.



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Test Objective for the Prone to Supine Upper Body Rolling Test:

This tests the ability to activate and sequence the local stabilizers for upper body extension pattern.

How to Perform the Prone to Supine Rolling Upper Body Test:

Instruct the patient to lie prone, on a flat surface with the legs extended and the arms flexed overhead. While keeping their lower body relaxed, they should reach their right arm up and reach across their body, while also extending their neck (following their hand with their eyes), to roll their body onto their back. Repeat using the left arm.

Functional Criteria:

Normal Prone to Supine Upper Body Rolling is the ability to roll from their stomach to their back, without using their lower body, in a controlled manner.

Additional Information:

Many times, gross stability dysfunctions combined with extension maneuvers can be identified using rolling patterns. A good sequence of core stability and segmental loading of the entire body is required to perform this test correctly.

Watch for simultaneous extension from the lower body, log rolling, and/or using flexion to complete the pattern.

Possible Findings:

- Functional: Ability to roll from the prone position to the supine position, no pain
 - » Indicates a higher postural stability/motor control dysfunction for upper body extension sequencing.
- Dysfunctional: Inability to roll from the prone position to the supine position, no pain
 - » Indicates a fundamental level stability/motor control dysfunction for upper body extension sequencing.

*If there is pain with the rolling pattern, consider it dysfunctional and remain in the Supported posture.



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Test Objective for the Prone to Supine Lower Body Rolling Test:

This tests the ability to activate and sequence the local stabilizers for lower body extension pattern.

How to Perform the Prone to Supine Rolling Lower Body Test:

Instruct the patient to lie supine, on a flat surface with the legs extended and the arms flexed overhead. While keeping their upper body relaxed, they should lift their right leg up across their body to roll onto their back. Repeat using the left leg..

Functional Criteria:

Normal Prone to Supine Lower Body Rolling is the ability to roll from their stomach to their back, without using their upper body, in a controlled manner.

Additional Information:

Many times, gross stability dysfunctions combined with extension maneuvers can be identified using rolling patterns. A good sequence of core stability and segmental loading of the entire body is required to perform this test correctly.

Watch for simultaneous extension from the lower body, log rolling, and/or using flexion to complete the pattern

Possible Findings:

- Functional: Able to roll from the prone to the supine position, no pain
 - » Indicates a higher postural stability/motor control dysfunction for lower body extension sequencing.
- Dysfunctional: Unable to roll from prone to the supine position, no pain
 - » Indicates a fundamental level stability/motor control dysfunction for lower body extension sequencing.

*If there is pain with the rolling pattern, consider it dysfunctional and remain in the Supported posture.

